

# Complemented Copies in $C(\beta\mathbb{N} \times \beta\mathbb{N})$ : Exact Convex Reductions, Positive Closure Principles, and a Separable Obstruction to the Cube-Averaging Route

Research note

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## Abstract

We study the problem of Plebanek, Rondoš, and Sobota asking whether  $C(\beta\mathbb{N} \times \beta\mathbb{N})$  contains a complemented isomorphic copy of  $C(K)$  for every separable compact space  $K$ . The problem is not resolved here, and no full solution or counterexample was located in a literature search through June 20, 2026. We prove, however, several exact reductions and closure results that substantially narrow the target.

First, the problem is equivalent to its restriction to separable compact convex spaces; in fact it is enough to consider the Bauer simplices  $\mathcal{P}(K)$  of probability measures on separable compact spaces  $K$ . It is also enough to consider the weak-star compact symmetric dual balls  $B_{\mathcal{M}(K)}$ . These reductions are quantitative and use norm-one projections. Second, every separable almost-Milyutin compactum gives a positive instance, hence so does every separable Dugundji compactum. Third, we isolate a concrete separable dyadic test space  $K_* = T^{\mathbb{N}}$  from a theorem of Blasco and Ivorra:  $K_*$  is not almost Milyutin, while the canonical averaging constants of the finite powers  $T^n$  tend to infinity. Thus no proof based solely on producing an averaging operator from a Cantor cube can settle the problem. Finally, we recast the question as a complementable-universality problem for  $\mathcal{K}(\ell_1, \ell_\infty)$  and prove a no-go lemma showing why the most direct operator-space projection cannot work.

## 1 The problem and notation

All compact spaces are Hausdorff and all Banach spaces are real. Write

$$Z = C(\beta\mathbb{N} \times \beta\mathbb{N}).$$

The source problem is the following.

**Problem 1.1** (Plebanek–Rondoš–Sobota [5]). Does  $Z$  contain a complemented isomorphic copy of  $C(K)$  for every separable compact space  $K$ ?

For Banach spaces  $X, Y$ , the notation  $X \xrightarrow{c} Y$  means that there are bounded operators

$$U : X \longrightarrow Y, \quad V : Y \longrightarrow X, \quad VU = I_X.$$

Thus  $U$  is an isomorphic embedding and  $UV$  is a bounded projection onto  $U[X]$ . We use the quantitative factorization constant

$$\gamma_Y(X) = \inf\{\|U\| \|V\| : U : X \rightarrow Y, V : Y \rightarrow X, VU = I_X\}, \tag{1}$$

with the convention that the infimum is  $+\infty$  if no such factorization exists.

The principal imported result from [5] is

$$C(2^\kappa) \xhookrightarrow{c} Z \quad (1 \leq \kappa \leq \mathfrak{c}). \quad (2)$$

Indeed, this is the second assertion of Corollary 1.4 there, applied to the two-point metric compactum. The same corollary also gives complemented copies of  $C([0, 1]^\kappa)$ .

## 2 Two elementary closure principles

We begin by recording the operator calculation behind the original partial result.

**Lemma 2.1** (Transitivity). *If  $X \xhookrightarrow{c} Y$  and  $Y \xhookrightarrow{c} W$ , then  $X \xhookrightarrow{c} W$ . Quantitatively,*

$$\gamma_W(X) \leq \gamma_Y(X) \gamma_W(Y).$$

*Proof.* Choose  $U_1 : X \rightarrow Y$ ,  $V_1 : Y \rightarrow X$ ,  $U_2 : Y \rightarrow W$ , and  $V_2 : W \rightarrow Y$  with  $V_i U_i = I$ . Then

$$(V_1 V_2)(U_2 U_1) = I_X,$$

and the norm estimate follows. In projection language, if  $A : Y \rightarrow W$  has complemented range with projection  $P$  and  $Q : Y \rightarrow Y$  projects onto a copy  $J[X]$ , then

$$\Pi = AQA^{-1}P$$

projects  $W$  onto  $AJ[X]$ . □

The next form is more flexible than an extension-operator argument.

**Proposition 2.2** (Averaging closure). *Let  $q : L \rightarrow K$  be a continuous surjection between compact spaces. Suppose there is a bounded operator*

$$A : C(L) \longrightarrow C(K) \quad \text{such that} \quad Aq^* = I_{C(K)}, \quad q^*f = f \circ q.$$

*If  $C(L) \xhookrightarrow{c} Z$ , then  $C(K) \xhookrightarrow{c} Z$ .*

*Proof.* The identity  $Aq^* = I$  says exactly that  $q^*C(K)$  is complemented in  $C(L)$ , with projection  $q^*A$ . Apply Lemma 2.1. Explicitly, if  $U : C(L) \rightarrow Z$  has complemented range and  $R : Z \rightarrow U[C(L)]$  is a projection, then

$$\Pi = Uq^*AU^{-1}R$$

projects  $Z$  onto  $Uq^*[C(K)]$ . □

A bounded operator  $A$  as above is called an *averaging operator* for  $q$ . If  $K$  is a closed subspace of a compact space  $L$  and the restriction map  $r : C(L) \rightarrow C(K)$  has a bounded right inverse  $E$ , then  $Er$  is a projection and the same conclusion follows. Thus the earlier extension-operator and retract cases are contained in Proposition 2.2.

### 3 An exact reduction to separable Bauer simplices

The following reduction is the main structural observation of this note. For a compact space  $K$ , let  $\mathcal{P}(K)$  be the weak-star compact space of probability measures on  $K$ .

**Theorem 3.1** (Canonical convexification). *For every compact space  $K$ , the Banach space  $C(K)$  is isometric to a norm-one complemented subspace of  $C(\mathcal{P}(K))$ .*

*More precisely, define*

$$\mathcal{A}_K : C(K) \longrightarrow C(\mathcal{P}(K)), \quad (\mathcal{A}_K f)(\mu) = \int_K f d\mu,$$

and

$$\mathcal{R}_K : C(\mathcal{P}(K)) \longrightarrow C(K), \quad (\mathcal{R}_K F)(x) = F(\delta_x).$$

Then  $\|\mathcal{A}_K\| = \|\mathcal{R}_K\| = 1$ ,  $\mathcal{R}_K \mathcal{A}_K = I_{C(K)}$ , and  $\mathcal{A}_K \mathcal{R}_K$  is a norm-one projection. Its range is the space of continuous affine functions on  $\mathcal{P}(K)$ .

*Proof.* Weak-star continuity of  $\mu \mapsto \mu(f)$  shows that  $\mathcal{A}_K$  is well defined. Since the Dirac measures belong to  $\mathcal{P}(K)$ ,

$$\|\mathcal{A}_K f\| = \sup_{\mu \in \mathcal{P}(K)} \left| \int f d\mu \right| = \sup_{x \in K} |f(x)| = \|f\|.$$

The Dirac map  $\delta : K \rightarrow \mathcal{P}(K)$  is continuous, so  $\mathcal{R}_K$  is a contraction. Moreover,

$$(\mathcal{R}_K \mathcal{A}_K f)(x) = \int_K f d\delta_x = f(x).$$

Hence  $\mathcal{A}_K \mathcal{R}_K$  is a norm-one projection onto  $\mathcal{A}_K[C(K)]$ .

Every function in that range is continuous and affine. Conversely, if  $F$  is continuous and affine on  $\mathcal{P}(K)$ , then for every finitely supported probability measure  $\mu = \sum_{j=1}^m a_j \delta_{x_j}$ ,

$$F(\mu) = \sum_{j=1}^m a_j F(\delta_{x_j}) = (\mathcal{A}_K \mathcal{R}_K F)(\mu).$$

Finitely supported probability measures are weak-star dense in  $\mathcal{P}(K)$ , so continuity gives  $F = \mathcal{A}_K \mathcal{R}_K F$  everywhere.  $\square$

**Lemma 3.2.** *If  $K$  is separable compact, then  $\mathcal{P}(K)$  is separable and  $w(\mathcal{P}(K)) \leq \mathfrak{c}$ .*

*Proof.* Let  $D \subseteq K$  be countable and dense. The set of probability measures of the form

$$\sum_{j=1}^m q_j \delta_{d_j}, \quad d_j \in D, \quad q_j \in \mathbb{Q} \cap [0, 1], \quad \sum_{j=1}^m q_j = 1,$$

is countable and weak-star dense in  $\mathcal{P}(K)$ . To verify density, take a basic weak-star neighbourhood determined by finitely many functions  $f_1, \dots, f_r$ . The map  $x \mapsto (f_1(x), \dots, f_r(x))$  has compact range in  $\mathbb{R}^r$ ; its integral against a probability measure can be approximated by a finite convex combination of values on the dense set  $D$ , and the coefficients can then be made rational.

Since every member of  $C(K)$  is determined by its restriction to  $D$ , one has  $|C(K)| \leq |\mathbb{R}^D| = \mathfrak{c}$ . The weak-star topology of  $\mathcal{P}(K)$  is generated by the coordinate maps  $\mu \mapsto \mu(f)$  for  $f \in C(K)$ . Consequently  $w(\mathcal{P}(K)) \leq \mathfrak{c}$ .  $\square$

Recall that  $\mathcal{P}(K)$  is a Bauer simplex: its extreme points are precisely the Dirac measures, the set  $\text{ext } \mathcal{P}(K) = \{\delta_x : x \in K\}$  is closed, and each  $\mu \in \mathcal{P}(K)$  has the unique representing probability measure  $\delta_*\mu$  on this extreme boundary.

**Theorem 3.3** (Exact convex reduction). *The following assertions are equivalent.*

- (i)  $C(K) \xrightarrow{c} Z$  for every separable compact space  $K$ .
- (ii)  $C(S) \xrightarrow{c} Z$  for every separable compact convex space  $S$ .
- (iii)  $C(\mathcal{P}(K)) \xrightarrow{c} Z$  for every separable compact space  $K$ .
- (iv)  $C(B_{\mathcal{M}(K)}) \xrightarrow{c} Z$  for every separable compact space  $K$ , where  $B_{\mathcal{M}(K)}$  is the weak-star compact unit ball of  $\mathcal{M}(K) = C(K)^*$ .

Thus, if a counterexample exists, one may choose a separable Bauer simplex, or a separable weak-star compact symmetric convex dual ball, as a counterexample.

*Proof.* The implications (i) $\Rightarrow$ (ii), (i) $\Rightarrow$ (iii), and (i) $\Rightarrow$ (iv) are immediate, because all spaces appearing in (ii)–(iv) are separable compact when  $K$  is separable. The assertion for  $\mathcal{P}(K)$  follows from Lemma 3.2. For the dual ball, weak-star separability follows because the countable set of finite rational convex combinations of the measures  $\pm\delta_d$ ,  $d \in D$ , is weak-star dense in  $B_{\mathcal{M}(K)}$ ; here we use that the weak-star closed convex hull of  $\{\pm\delta_x : x \in K\}$  is the whole dual unit ball. The same coordinate argument as in Lemma 3.2 gives weight at most  $\mathfrak{c}$ .

For (iii) $\Rightarrow$ (i), Theorem 3.1 gives  $C(K) \xrightarrow{c} C(\mathcal{P}(K))$ , and transitivity finishes the proof.

For (iv) $\Rightarrow$ (i), define

$$J_K : C(K) \rightarrow C(B_{\mathcal{M}(K)}), \quad (J_K f)(\mu) = \mu(f),$$

and

$$S_K : C(B_{\mathcal{M}(K)}) \rightarrow C(K), \quad (S_K F)(x) = F(\delta_x).$$

Then  $J_K$  is an isometry,  $\|S_K\| = 1$ , and  $S_K J_K = I$ . Hence  $C(K)$  is norm-one complemented in  $C(B_{\mathcal{M}(K)})$ , and transitivity again applies.  $\square$

*Remark 3.4.* Theorem 3.3 rules out a common simplification: passing to probability measures does not merely enlarge the positive class; it is an exact reformulation of the full problem. The remaining difficulty is already present for separable Bauer simplices.

## 4 The almost-Milyutin positive class

A compact space  $K$  is called *almost Milyutin* if there exist a Cantor cube  $2^\Gamma$ , a continuous surjection  $q : 2^\Gamma \rightarrow K$ , and a bounded averaging operator  $A : C(2^\Gamma) \rightarrow C(K)$  for  $q$ . If  $A$  can be chosen regular and positive, one obtains the usual Milyutin property.

**Lemma 4.1** (Coordinate reduction). *Suppose  $q : 2^\Gamma \rightarrow K$  is a continuous surjection onto a compact space of weight  $\kappa$  and  $q$  admits a bounded averaging operator. Then there are  $\Lambda \subseteq \Gamma$  with  $|\Lambda| \leq \kappa$ , a continuous surjection  $q_\Lambda : 2^\Lambda \rightarrow K$ , and a bounded averaging operator for  $q_\Lambda$  of no larger norm.*

*Proof.* Embed  $K$  into  $[0, 1]^\kappa$ . Each scalar coordinate of the composite  $2^\Gamma \rightarrow K \rightarrow [0, 1]^\kappa$  depends on only countably many coordinates of  $2^\Gamma$ . Taking the union of these coordinate sets gives  $\Lambda \subseteq \Gamma$  with  $|\Lambda| \leq \kappa$  such that  $q = q_\Lambda \circ \pi_\Lambda$  for some onto map  $q_\Lambda : 2^\Lambda \rightarrow K$ .

If  $A$  averages  $q$ , define

$$A_\Lambda g = A(g \circ \pi_\Lambda), \quad g \in C(2^\Lambda).$$

Then  $\|A_\Lambda\| \leq \|A\|$  and  $A_\Lambda q_\Lambda^* = I_{C(K)}$ . □

**Theorem 4.2.** *If  $K$  is a separable almost-Milyutin compact space, then  $C(K) \xrightarrow{c} Z$ . In particular, the answer to Problem 1 is positive for every separable Dugundji compact space.*

*Proof.* A separable compact space has weight at most  $\mathfrak{c}$ . By Lemma 4.1, the almost-Milyutin witness can be chosen with  $|\Gamma| \leq \mathfrak{c}$ . Hence

$$C(K) \xrightarrow{c} C(2^\Gamma) \xrightarrow{c} Z,$$

where the first factorization comes from the averaging operator and the second from (2).

Haydon proved that every Dugundji compactum is Milyutin [3]; therefore the final assertion follows. □

This theorem strictly extends the elementary retract and extension-operator subcases, but it still does not cover all separable compact spaces, as the next section shows.

## 5 A separable dyadic obstruction to the cube route

Blasco and Ivorra constructed a particularly sharp obstruction to bounded averaging operators from Cantor cubes [1]. We recall the construction in a form tailored to the present problem.

Let  $\alpha = \omega_1$  and let  $T$  be the quotient of  $2^\alpha$  obtained by identifying two distinct points. For a compact space  $L$ , put

$$p(L) = \inf \{ \|A\| : q : 2^\Gamma \twoheadrightarrow L, A : C(2^\Gamma) \rightarrow C(L), Aq^* = I_{C(L)} \}, \quad (3)$$

with  $p(L) = +\infty$  if no such pair exists.

**Theorem 5.1** (Blasco–Ivorra). *For every positive integer  $n$ ,*

$$p(T^n) \geq n + 1,$$

*although each finite power  $T^n$  is almost Milyutin. Moreover,*

$$K_* = T^\mathbb{N}$$

*is dyadic but not almost Milyutin; equivalently,  $p(K_*) = +\infty$ .*

The space  $K_*$  is a legitimate test object for Problem 1.

**Proposition 5.2.** *The compact space  $K_*$  is separable, dyadic, and has weight at most  $\mathfrak{c}$ .*

*Proof.* The Hewitt–Marczewski–Pondiczery theorem implies that  $2^{\omega_1}$  is separable, since  $\omega_1 \leq \mathfrak{c}$ ; see Ross and Stone [7]. Thus its quotient  $T$  is separable, and a countable product of separable spaces is separable. If  $q : 2^\alpha \rightarrow T$  is the quotient map, then

$$q^\mathbb{N} : (2^\alpha)^\mathbb{N} \longrightarrow T^\mathbb{N}$$

is onto, while  $(2^\alpha)^\mathbb{N} \cong 2^{\alpha \times \mathbb{N}} \cong 2^\alpha$ . Hence  $K_*$  is dyadic. The weight estimate is immediate. □

Consequently, a proof of Problem 1 cannot proceed by showing that every separable compact space is almost Milyutin. The gap between canonical averaging and arbitrary isomorphic complementability is essential. The following quantitative observation makes this gap precise.

**Proposition 5.3** (Uniform finite-stage test). *If  $C(K_*) \xrightarrow{c} Z$ , then*

$$\sup_{n \geq 1} \gamma_Z(C(T^n)) < \infty.$$

*More precisely,*

$$\gamma_Z(C(T^n)) \leq \gamma_Z(C(K_*)) \quad (n \geq 1).$$

*Proof.* Fix  $t_0 \in T$ . The coordinate projection  $\pi_n : K_* \rightarrow T^n$  induces an isometric embedding

$$J_n : C(T^n) \rightarrow C(K_*), \quad J_n f = f \circ \pi_n.$$

The map

$$R_n : C(K_*) \rightarrow C(T^n), \quad R_n g(x_1, \dots, x_n) = g(x_1, \dots, x_n, t_0, t_0, \dots)$$

is a contraction and  $R_n J_n = I$ . Composing any factorization of  $I_{C(K_*)}$  through  $Z$  with  $J_n$  and  $R_n$  gives the required factorization for  $C(T^n)$  with no larger norm product.  $\square$

Combining Theorem 5.1 and Proposition 5.3, we obtain a concrete possible route to a negative answer: it would suffice to prove, for the finite powers  $T^n$ , a bound of the form

$$p(T^n) \leq \Phi(\gamma_Z(C(T^n))) \tag{4}$$

for one finite-valued function  $\Phi$  independent of  $n$ . Indeed, the right-hand side would remain bounded if  $C(K_*) \xrightarrow{c} Z$ , while the left-hand side is at least  $n+1$ . No estimate such as (4) is currently available: the Blasco–Ivorra constant concerns canonical pullback copies arising from continuous maps, whereas  $\gamma_Z$  allows arbitrary isomorphic copies. This is the precise point at which the candidate stops short of a counterexample.

## 6 The compact-operator reformulation

There are natural isometric identifications

$$\begin{aligned} Z &\cong C(\beta\mathbb{N}, C(\beta\mathbb{N})) \cong C(\beta\mathbb{N}, \ell_\infty) \\ &\cong \mathcal{K}(\ell_1, \ell_\infty) \cong \ell_\infty \hat{\otimes}_\varepsilon \ell_\infty. \end{aligned} \tag{5}$$

For the middle identification, an operator  $S : \ell_1 \rightarrow \ell_\infty$  is represented by the sequence  $(Se_n)_{n \in \mathbb{N}}$ . It is compact exactly when this sequence has relatively norm-compact range; such a sequence extends uniquely to a norm-continuous map from  $\beta\mathbb{N}$  into  $\ell_\infty$ .

If  $K$  is separable and  $(d_n)$  is dense in  $K$ , then

$$E_K : C(K) \longrightarrow \ell_\infty, \quad E_K f = (f(d_n))_{n \in \mathbb{N}}, \tag{6}$$

is an isometry. Thus Problem 1 asks whether  $\mathcal{K}(\ell_1, \ell_\infty)$  is complementably universal for those  $C(K)$ -spaces that admit a countable 1-norming set.

A tempting strategy is to place  $\mathcal{K}(\ell_1, E_K C(K))$  inside  $\mathcal{K}(\ell_1, \ell_\infty)$  and try to project onto it. The next lemma shows that this strategy cannot give anything new.

**Lemma 6.1** (Operator-space no-go lemma). *Let  $X$  be a closed subspace of a Banach space  $Y$ . If  $\mathcal{K}(\ell_1, X)$  is complemented in  $\mathcal{K}(\ell_1, Y)$ , then  $X$  is complemented in  $Y$ . Moreover, a projection of norm  $M$  in the operator space produces a projection of norm at most  $M$  in  $Y$ .*

*Proof.* For  $y \in Y$ , define the rank-one operator

$$R_y : \ell_1 \rightarrow Y, \quad R_y(a) = a_1 y.$$

Suppose  $Q : \mathcal{K}(\ell_1, Y) \rightarrow \mathcal{K}(\ell_1, X)$  is a projection. Set

$$P(y) = Q(R_y)e_1.$$

Then  $P : Y \rightarrow X$  is bounded with  $\|P\| \leq \|Q\|$ . If  $x \in X$ , then  $R_x \in \mathcal{K}(\ell_1, X)$ , so  $Q(R_x) = R_x$  and  $P(x) = x$ . Hence  $P$  is a projection onto  $X$ .  $\square$

Therefore a positive solution must use a genuinely twisted copy of  $C(K)$  inside  $\mathcal{K}(\ell_1, \ell_\infty)$ ; it cannot simply project all compact operators into the evaluation copy of  $C(K)$  inside  $\ell_\infty$ .

## 7 A sufficient coupling mechanism

The source paper succeeds for compact groups by constructing continuous families of probability couplings supported on the fibres of a difference map. The abstract mechanism is short enough to state separately.

**Proposition 7.1** (Fibre-supported coupling criterion). *Let  $\rho : L \rightarrow K$  be a continuous surjection between compact spaces. Suppose there is a weak-star continuous map*

$$K \longrightarrow \mathcal{P}(L), \quad x \longmapsto \nu_x,$$

*such that  $\text{supp } \nu_x \subseteq \rho^{-1}(x)$  for every  $x \in K$ . Then*

$$Af(x) = \int_L f d\nu_x$$

*defines a norm-one positive averaging operator for  $\rho$ . Consequently, if  $C(L) \xhookrightarrow{c} Z$ , then  $C(K) \xhookrightarrow{c} Z$ .*

*Proof.* Weak-star continuity gives  $Af \in C(K)$  and  $\|A\| = 1$ . For  $g \in C(K)$ ,

$$A(g \circ \rho)(x) = \int_{\rho^{-1}(x)} g(\rho(y)) d\nu_x(y) = g(x).$$

Apply Proposition 2.2.  $\square$

A full positive solution would follow, for example, from a theorem that every separable compact space is the target of such a map from a compact  $L$  whose  $C(L)$  is already known to be complemented in  $Z$ . The Blasco-Ivorra example shows that one cannot always take  $L$  to be a Cantor cube. The compact-group construction in [5] supplies exactly this criterion in a highly structured setting, but no general replacement for the group operation is presently known.

## 8 Status of the literature and conclusion

The exact wording of Problem 1, the title of [5], its citation trail, and combinations of the terms “ $C(\beta\mathbb{N} \times \beta\mathbb{N})$ ”, “separable compact”, “complemented”, “Milyutin”, and “averaging operator” were searched through June 20, 2026. No paper giving a complete affirmative answer or a counterexample was located. The recent preprint of Rondoš and Sobota [6] gives a powerful characterization of complemented copies  $C(L)$  when  $L$  is metrizable, hence when  $C(L)$  is separable as a Banach space. It does not cover the nonmetrizable separable compacta that form the unresolved part of Problem 1.

The strongest rigorous conclusions obtained here are:

1. The original question is *exactly equivalent* to the same question for separable Bauer simplices  $\mathcal{P}(K)$ , and also to the same question for separable symmetric weak-star dual balls  $B_{\mathcal{M}(K)}$ .
2. Every separable almost-Milyutin compactum, in particular every separable Dugundji compactum, gives a positive instance.
3. The separable dyadic compactum  $K_* = T^{\mathbb{N}}$  is not almost Milyutin and therefore blocks every universal proof based on a bounded averaging from a Cantor cube. Its finite powers provide the quantitative test  $p(T^n) \geq n + 1$ .
4. In the operator model  $Z \cong \mathcal{K}(\ell_1, \ell_\infty)$ , the obvious attempt to project onto  $\mathcal{K}(\ell_1, C(K))$  is impossible unless the original evaluation copy of  $C(K)$  was already complemented in  $\ell_\infty$ .

Thus the full problem remains open in this note. The two sharpest next targets are either to prove that the Blasco–Ivorra space  $K_*$  cannot factor through  $Z$ , perhaps by establishing a comparison such as (4), or to build a noncanonical twisted factorization of every separable Bauer simplex through  $\mathcal{K}(\ell_1, \ell_\infty)$ .

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